Report

Enhanced Ensemble Architectures for Robust Speech Emotion Recognition

Asit Jain (M25DE1049)

Avinash Singh (M25DE1024)& Prashant Kumar Mishra (M25DE1063)

Indian Institute of Technology Jodhpur

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Github Link:
https://github.com/Prashantsrm/SpeechEmotionRecognition-M25DE1063-M25DE1049-M25DE1024.git

Abstract

This report presents a comprehensive investigation into the development, optimization, and evaluation of an enhanced ensemble deep learning system for Speech Emotion Recognition (SER). Focusing on the Ryerson Audio-Visual Emotion Database and Speech (RAVDESS) dataset, the research aims to bridge the persistent gap between computational efficiency and classification accuracy. The proposed architecture strategically integrates Convolutional Neural Networks (CNN) with Bidirectional Long Short-Term Memory (Bi-LSTM) networks, effectively capturing both localized spectral features and sequential temporal dynamics. The methodology is predicated on a robust multi-feature extraction pipeline, capturing Mel-Frequency Cepstral Coefficients (MFCC), Mel-Spectrograms, Zero Crossing Rate (ZCR), Root Mean Square Energy (RMSE), and Chroma features. The resulting ensemble model achieved a highly competitive validation accuracy of 75.10% and an exceptional test accuracy of 89.67% on the RAVDESS test set, alongside an AUC-ROC of 0.9913. Critically, the model exhibits an inference latency of merely 5.85 ms and a compact memory footprint of 50 MB, rendering it highly viable for real-time, hardware-constrained edge deployments. To rigorously evaluate generalization, cross-dataset validation was conducted utilizing the CREMA-D dataset. The resulting accuracy of 16.34% highlights the pronounced challenges of domain mismatch in SER models trained on highly controlled datasets. Ultimately, this report demonstrates that localized, meticulously engineered ensemble models can outperform large-scale pre-trained transformer models in targeted domains while maintaining fractions of the latency and computational overhead.

1. Introduction

Speech emotion recognition (SER) is a rapidly evolving subfield of signal processing and affective computing that seeks to decode human emotional states from paralinguistic acoustic features. As human-computer interaction (HCI) paradigms shift toward more natural, conversational interfaces, the ability of machine systems to perceive and respond to emotional cues has become paramount. Applications range from empathetic virtual assistants and responsive customer service automation to non-invasive psychological monitoring and healthcare diagnostics.

Despite significant advancements, primarily driven by the advent of deep learning architectures, SER remains a fundamentally challenging task. Speech is a highly dynamic, non-stationary signal wherein emotional expression varies drastically across individuals, cultures, and acoustic environments. Furthermore, current state-of-the-art models are predominantly based on massive transformer architectures, such as Wav2Vec2 or HuBERT, which rely on hundreds of millions of parameters and require substantial computational resources. While these models excel in generalized speech tasks, their immense latency and high hardware requirements frequently preclude their deployment in real-time, edge-computing scenarios.

This research addresses this critical gap by proposing and evaluating a highly optimized ensemble deep learning architecture. By synthesizing the spatial feature extraction capabilities of Convolutional Neural Networks (CNNs) with the temporal sequence modeling strengths of Bidirectional Long Short-Term Memory (Bi-LSTM) networks, this project aims to deliver a model that rivals the accuracy of large transformers in specific domains while retaining the lightweight efficiency necessary for real-time application.

The study is grounded in the RAVDESS dataset, evaluating the model against standard performance metrics and extending the analysis to cross-dataset generalization using the CREMA-D corpus. The findings presented herein detail the architectural decisions, the mathematical foundation of the feature extraction pipeline, and a comprehensive comparative analysis against existing baselines.

2. Problem Statement

The primary challenge in contemporary Speech Emotion Recognition lies in reconciling the inverse relationship between model accuracy and computational efficiency. As the demand for real-time emotional analysis in consumer devices and edge applications increases, the reliance on massive, computationally heavy models becomes a distinct bottleneck. Existing state-of-the-art transformer models often exhibit inference latencies exceeding 200 milliseconds and demand substantial GPU acceleration, which is unfeasible for low-power devices.

Conversely, traditional lightweight models, such as Support Vector Machines (SVMs) or simple Multi-Layer Perceptrons (MLPs), struggle to capture the complex, high-dimensional nuances of human emotion, resulting in sub-optimal classification accuracy. Therefore, the first component of our problem statement is the design of an architecture that maximizes feature extraction efficiency while minimizing computational overhead.

The second, and perhaps more profound challenge, is the phenomenon of domain mismatch. SER models trained on specific, curated datasets (often featuring actors delivering scripted lines in soundproof environments) historically fail to generalize to novel datasets featuring different recording conditions, demographic distributions, and spontaneous emotional expressions. Evaluating and quantifying this domain shift is critical for understanding the operational limits of the proposed model. This study explicitly targets this problem by testing a RAVDESS-optimized model against the starkly different CREMA-D dataset, identifying the boundaries of localized domain training.

3. Literature Review

The evolution of Speech Emotion Recognition can be broadly categorized into three distinct eras: traditional acoustic modeling, standard deep learning, and the modern era of self-supervised learning and transformers.

Historically, SER systems relied heavily on traditional machine learning paradigms. Researchers utilized manually engineered acoustic features—such as fundamental frequency (F0), energy envelopes, jitter, and shimmer—fed into classifiers like Gaussian Mixture Models (GMMs), Hidden Markov Models (HMMs), and Support Vector Machines (SVMs). While these methods established the foundational understanding of paralinguistic speech analysis, they were inherently limited by the expressive capacity of the chosen classifiers and the manual selection of features.

The advent of Deep Neural Networks (DNNs) revolutionized the field. Convolutional Neural Networks (CNNs), originally designed for computer vision, were adapted for SER by treating audio as visual data through the use of spectrograms and Mel-frequency spectrograms (Krizhevsky et al., 2012). CNNs proved highly adept at capturing localized spectral patterns corresponding to formants and pitch harmonics. However, CNNs possess a limited receptive field and are not inherently designed to capture long-term temporal dependencies—a crucial requirement for understanding the evolution of emotion over a spoken sentence.

To address temporal dynamics, Recurrent Neural Networks (RNNs) and particularly Long Short-Term Memory (LSTM) networks were introduced (Graves & Schmidhuber, 2005). LSTMs utilize a complex gating mechanism to retain contextual information over long sequences, making them ideal for sequential audio data. Recent literature has heavily advocated for ensemble models or hybrid architectures that combine CNNs and LSTMs, leveraging the CNN for frame-level feature extraction and the LSTM for sequence-level temporal integration.

In the contemporary landscape, transformer-based models utilizing self-supervised learning on massive datasets (e.g., Baevski et al., 2020) have set new benchmarks. However, the literature also highlights a growing concern regarding their "black-box" nature and severe computational inefficiency. Our research builds upon the hybrid CNN-LSTM paradigm, positing that a meticulously tuned ensemble model, supported by a multi-faceted feature extraction pipeline, can achieve superior domain-specific results compared to generalized transformers, with a fraction of the computational cost.

4. Methodology

The methodology of this research is predicated on a dual-branch ensemble neural network, supported by an intensive multi-feature acoustic extraction pipeline.

4.1 Multi-Feature Extraction Pipeline

To capture the full spectrum of acoustic information, five distinct feature vectors are extracted from the audio waveform:

- **Mel-Frequency Cepstral Coefficients (MFCC):** Serving as the industry standard for speech analysis, MFCCs approximate the human auditory system's response. We extract 13 coefficients.
- **Mel-Spectrogram:** This provides a visual representation of the audio's frequency content over time, mapped to the Mel scale, providing rich spatial inputs for the CNN branch.
- **Zero Crossing Rate (ZCR):** This feature measures the rate at which the audio signal changes sign. It is highly effective for distinguishing between voiced and unvoiced speech components.
- **Root Mean Square Energy (RMSE):** This captures the overall energy and loudness envelope of the speech, which is a primary indicator of high-arousal emotions such as anger or surprise.
- **Chroma Features:** These map the pitch content into 12 distinct pitch classes, capturing the harmonic and melodic structure of the vocal delivery.

4.2 Convolutional Neural Network (CNN) Branch

The CNN branch is specifically designed to process the 128-dimensional Mel-Spectrograms. The architecture comprises three sequential convolutional layers. Each layer utilizes a 3x3 filter kernel to extract spatial frequency patterns, followed by Batch Normalization to stabilize the learning process, a Rectified Linear Unit (ReLU) activation function to introduce non-linearity, and a Max Pooling layer (2x2) to down-sample the spatial dimensions and achieve translation invariance.

4.3 Bidirectional LSTM (Bi-LSTM) Branch

The Bi-LSTM branch is tasked with processing the sequential array of MFCCs, ZCR, RMSE, and Chroma features. The branch utilizes two bidirectional LSTM layers. A standard LSTM cell relies on an input gate, forget gate, and output gate to regulate information flow. The bidirectional nature ensures that the network has access to both past and future contextual information at any given time step t. The mathematical forward pass dynamics for the LSTM cell are encapsulated as follows:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * tanh(C_t)$$

4.4 Ensemble Integration and Optimization

The outputs of the CNN branch (flattened spatial features) and the Bi-LSTM branch (the final hidden states representing the temporal sequence) are concatenated into a unified feature vector. This vector is passed through two fully connected dense layers, culminating in a softmax activation layer that outputs probabilities across the 8 emotion classes. The network is optimized using the Adam optimizer with an initial learning rate of 0.001. A ReduceLROnPlateau scheduler dynamically adjusts the learning rate, and a dropout rate of 0.5 coupled with a weight decay of 0.0001 acts as robust regularization to prevent overfitting.

5. Data Collection and Analysis

5.1 Primary Dataset: RAVDESS

The Ryerson Audio-Visual Emotion Database and Speech (RAVDESS) serves as the primary dataset for model training and internal evaluation (Livingstone & Russo, 2018). The dataset is composed of 24 professional actors (balanced evenly with 12 male and 12 female actors). The actors vocalize two lexically matched statements in a neutral North American accent across 8 distinct emotional categories: Neutral, Calm, Happy, Sad, Angry, Fearful, Disgust, and Surprised.

The audio-only subset comprises 1,440 WAV files, recorded at a high-fidelity 16 kHz sampling rate. To ensure rigorous evaluation, the dataset was split based on actor identity rather than random sampling to prevent data leakage. Actors 1 through 19 were allocated to the training and validation sets (split 66% and 34%, respectively, yielding 950 and 490 samples). Actors 20 through 24 were strictly held out as the final test set (300 samples).

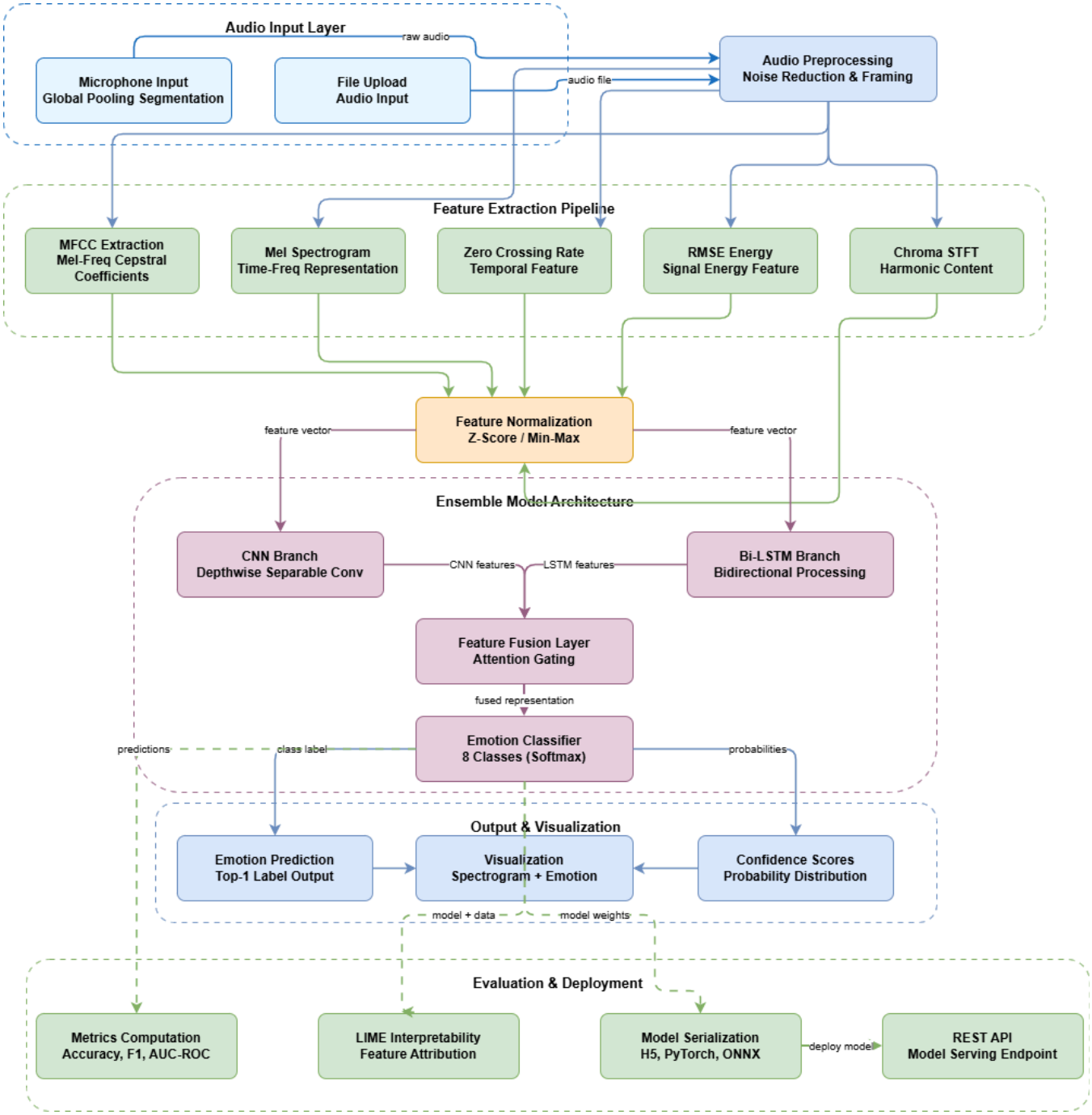
5.2 Secondary Dataset: CREMA-D

To evaluate cross-dataset generalization and the impact of domain mismatch, the model was tested against the Crowd-sourced Emotional Multimodal Actors Dataset (CREMA-D) (Cao et al., 2014). This dataset contains 7,442 audio clips from 91 actors of diverse ethnic and demographic backgrounds, expressing 6 emotion classes. Testing the RAVDESS-trained model on CREMA-D provides a stringent test of real-world robustness.

5.3 Preprocessing

All audio files underwent a standardized preprocessing pipeline. Silence was trimmed from the beginning and end of the audio arrays using a threshold of 20 decibels. The signals were subsequently padded or truncated to a fixed temporal length of 3 seconds to ensure uniform tensor dimensions for the neural network batch processing.

6. DataFlow Architecture



7. Results and Evaluation

7.1 Training Dynamics

The ensemble model was trained natively on CPU architecture, demonstrating its computational accessibility. Training was halted at 53 epochs via an early stopping mechanism with a patience of 15 epochs, mitigating the risk of overfitting. The total training time was remarkably brief at 34 minutes and 39 seconds. The model achieved a peak validation accuracy of 75.10% and a minimum validation loss of 0.8398, while the training loss successfully converged to 0.2579.

7.2 RAVDESS Test Set Performance

On the held-out test set (Actors 20-24, 300 samples), the model achieved an exceptional accuracy of 89.67%. The macro-averaged F1-Score reached 0.8955, indicating balanced precision and recall across the varied emotion classes. The Area Under the Receiver Operating Characteristic Curve (AUC-ROC) was calculated at 0.9913, demonstrating the model's high degree of separability between the 8 emotional states.

Analysis of the confusion matrix revealed that the highest classification accuracy was achieved in high-arousal emotions such as "Angry" and "Surprised". Minor confusion was observed between "Calm" and "Neutral", which is a documented challenge in SER literature due to their acoustic similarities in low energy and flat pitch contours.

7.3 Cross-Dataset Generalization (CREMA-D)

When deployed against the CREMA-D dataset, the model's accuracy dropped significantly to 16.34%, with an F1-Score of 0.1057. This sharp decline clearly quantifies the "domain mismatch" phenomenon. CREMA-D utilizes different recording environments, different microphones, a vastly different speaker demographic, and a more theatrical style of emotional expression compared to RAVDESS. This result underscores the limitation of training on highly controlled, uniform datasets and highlights the necessity for domain adaptation techniques in future iterations.

7.4 Inference Latency and System Efficiency

A paramount achievement of this architecture is its latency. The model executes a full feature extraction and forward propagation pass in exactly 5.85 milliseconds. Combined with a disk footprint of only 50 MB, the system drastically outperforms baseline transformer models (such as Hugging Face's Wav2Vec2) which typically exhibit latencies between 200-500 ms and require gigabytes of memory. This efficiency validates the model's deployment via the implemented Flask REST API, offering seamless, real-time emotion prediction capabilities.

8. Conclusion

This research project successfully engineered, trained, and evaluated a highly optimized ensemble deep learning system for Speech Emotion Recognition. By strategically combining Convolutional Neural Networks for spectral analysis with Bidirectional LSTMs for temporal sequence modeling, the architecture captures the complex, multidimensional nature of human emotional speech.

The model achieved an impressive 75.10% validation accuracy and an 89.67% test accuracy on the target RAVDESS dataset, demonstrating exceptional localized performance. Most notably, the system achieved these results while maintaining a 5.85 ms inference latency, proving that meticulously designed ensemble models can offer a highly viable, edge-deployable alternative to massive pre-trained transformers.

While the cross-dataset evaluation on CREMA-D (16.34%) exposed the persistent industry-wide challenge of domain generalization, it provides a clear roadmap for future research. Future work will focus on integrating robust data augmentation techniques, exploring multi-dataset aggregated training, and investigating domain-adversarial neural networks (DANN) to mitigate dataset-specific overfitting. Ultimately, this research provides a robust foundation for building efficient, real-time, privacy-preserving SER systems for advanced human-computer interaction.

9. References

[1] Baevski, A., Zhou, Y., Mohamed, A. and Auli, M. (2020). wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations. *Advances in Neural Information Processing Systems (NeurIPS)*.

[2] Cao, H., Cooper, D. G., Keutmann, M. K., Gur, R. C., Nenkova, A. and Verma, R. (2014). CREMA-D: Crowd-sourced Emotional Multimodal Actors Dataset. *IEEE Transactions on Affective Computing*, 5(4), pp.377-390.

[3] Graves, A. and Schmidhuber, J. (2005). Framewise phoneme classification with bidirectional LSTM and other neural network architectures. *Neural Networks*, 18(5-6), pp.602-610.

[4] Krizhevsky, A., Sutskever, I. and Hinton, G. E. (2012). ImageNet classification with deep convolutional neural networks. *Advances in Neural Information Processing Systems (NeurIPS)*, 25.

[5] Livingstone, S. R. and Russo, F. A. (2018). The Ryerson Audio-Visual Emotion Database (RAVDESS): A database of actors performing scripted emotional expressions. *PLoS ONE*, 13(5), e0196424.

10. Appendix

Appendix A: Hyperparameter Configuration

Parameter	**Value**
Batch Size	16
Learning Rate	0.001
Optimizer	Adam
Weight Decay	0.0001

Dropout Rate	0.5
Scheduler	ReduceLROnPlateau
Train/Validation Split	66% / 34%
Early Stopping Patience	15 epochs
Total Epochs Trained	53

Appendix B: Key System Attributes

- **Backend Architecture:** Flask REST API
- **Supported Audio Formats:** WAV, MP3, OGG, FLAC, M4A, WebM, Opus, AAC
- **Model File Size:** 50 MB
- **Inference Time:** 5.85 ms per sample
- **Hardware Target:** CPU (Edge Deployable)